

Total No. of Questions : 8]

SEAT No. :

PE2749

[Total No. of Pages : 2

[6583]-332

T.E. (Computer Engineering) (Honors)
DATA SCIENCE AND VISUALIZATION
(2019 Pattern) (Semester - V) (310501)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** What is Linear regression? List the applications where linear regression can be applied? **[6]**
- b)** State and explain how Naïve Bayes classifier can be used to solve the classification problems? **[6]**
- c)** Write a note on association rules. **[6]**

OR

- Q2) a)** What is clustering? Explain K-means clustering algorithm. **[6]**
- b)** Explain Apriori Algorithm used in machine learning with valid example. **[6]**
- c)** Illustrate how will you evaluate association rules. **[6]**

- Q3) a)** State and explain the different constituents of the decision tree. **[9]**
- b)** Write a note on the perceptron model. **[8]**

OR

- Q4) a)** When do you use backpropagation in Neural Networks? Explain by taking a suitable example. **[9]**
- b)** What is entropy? How entropy is calculated explain with a suitable example. **[8]**

- Q5) a)** Define the term Dashboard along with its evolution and steps to design the dashboard. **[9]**

- b)** Write a note on : **[9]**
- i) Pie charts
 - ii) Bar graphs
 - iii) Scatter plots

OR

P.T.O.

- Q6)** a) Explain the terms Network hierarchies and reports associated with data visualization. [6]
- b) Write a note on advanced visualization techniques and explain anyone of them. [6]
- c) Write a note on 'display media for Dashboard'. [6]
- Q7)** a) What are different types of data model explain in brief. [6]
- b) List the advantages of multi-dimensional data model? [6]
- c) Discuss the challenges of clustering High-Dimensional data. [5]

OR

- Q8)** a) Explain the need of data modelling. [6]
- b) Explain multidimensional data model with one example. [6]
- c) What do you mean by Principal Component Analysis? [5]

